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March 20, 2017

GO2-17-067

10 CFR 50.73

U.S. Nuclear Regulatory Commission
ATTN: Document Control Desk
Washington, D.C. 20555-0001

Subject: **COLUMBIA GENERATING STATION, DOCKET NO. 50-397**
LICENSEE EVENT REPORT NO. 2017-001-00

Dear Sir or Madam:

Transmitted herewith is Licensee Event Report No. 2017-001-00 for Columbia Generating Station. This report is submitted pursuant to 10 CFR 50.73(a)(2)(v)(D).

There are no commitments being made to the Nuclear Regulatory Commission by this letter. If you have any questions or require additional information, please contact Ms. D.M. Wolfgramm, Regulatory Compliance Supervisor, at (509) 377-4792.

Executed on this 20th day of March, 2017.

Respectfully,

A. L. Javorik
Vice President, Engineering

Enclosure: Licensee Event Report 2017-001-00

cc: NRC Region IV Administrator
NRC NRR Project Manager
NRC Senior Resident Inspector/988C
CD Sonoda – BPA/1399
WA Horin – Winston & Strawn

NRC FORM 366 (06-2016) U.S. NUCLEAR REGULATORY COMMISSION	APPROVED BY OMB: NO. 3150-0104 EXPIRES: 10/31/2018
LICENSEE EVENT REPORT (LER) (See Page 2 for required number of digits/characters for each block) (See NUREG-1022, R.3 for instruction and guidance for completing this form http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/)	Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Privacy and Information Collections Branch (T-5 F53), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollects.Resource@nrc.gov, and to the Desk Officer, Office of Information and Regulatory Affairs, NEOB-10202, (3150-0104), Office of Management and Budget, Washington, DC 20503. If a means used to impose an information collection does not display a currently valid OMB control number, the NRC may not conduct or sponsor, and a person is not required to respond to, the information collection.

1. FACILITY NAME	2. DOCKET NUMBER	3. PAGE
Columbia Generating Station	05000 397	1 OF 3

4. TITLE

Contactor Coil Failure Results in Tripping of HPCS Diesel Mixed Air Fan

5. EVENT DATE			6. LER NUMBER			7. REPORT DATE			8. OTHER FACILITIES INVOLVED	
MONTH	DAY	YEAR	YEAR	SEQUENTIAL NUMBER	REV NO.	MONTH	DAY	YEAR	FACILITY NAME	DOCKET NUMBER 05000
01	25	2017	2017 -	001 -	00	03	20	2017	FACILITY NAME	DOCKET NUMBER 05000

9. OPERATING MODE	11. THIS REPORT IS SUBMITTED PURSUANT TO THE REQUIREMENTS OF 10 CFR §: (Check all that apply)			
1	☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)
	☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)
	☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)
	☐ 20.2203(a)(2)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)
10. POWER LEVEL	☐ 20.2203(a)(2)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(v)(A)	☐ 73.71(a)(4)
	☐ 20.2203(a)(2)(iii)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(v)(B)	☐ 73.71(a)(5)
	☐ 20.2203(a)(2)(iv)	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(1)
100	☐ 20.2203(a)(2)(v)	☐ 50.73(a)(2)(i)(A)	☑ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(i)
	☐ 20.2203(a)(2)(vi)	☐ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(vii)	☐ 73.77(a)(2)(ii)
		☐ 50.73(a)(2)(i)(C)	☐ OTHER	Specify in Abstract below or in NRC Form 366A

12. LICENSEE CONTACT FOR THIS LER

LICENSEE CONTACT	TELEPHONE NUMBER (Include Area Code)
Tracey Parmelee, Principal Engineer, Compliance	(509) 377-8395

13. COMPLETE ONE LINE FOR EACH COMPONENT FAILURE DESCRIBED IN THIS REPORT

CAUSE	SYSTEM	COMPONENT	MANU-FACTURER	REPORTABLE TO EPIX	CAUSE	SYSTEM	COMPONENT	MANU-FACTURER	REPORTABLE TO EPIX
B	BG	CNTR	G080	Y					

14. SUPPLEMENTAL REPORT EXPECTED	15. EXPECTED SUBMISSION DATE	MONTH	DAY	YEAR
☐ YES (If yes, complete 15. EXPECTED SUBMISSION DATE) ☑ NO				

ABSTRACT (Limit to 1400 spaces, i.e., approximately 15 single-spaced typewritten lines)

On January 25, 2017 at 1836 PST, smoke was detected in the High Pressure Core Spray (HPCS) System diesel room with no indication of a fire. Immediate recovery actions by Operations personnel included opening the disconnect for the affected motor starter, at which point the smoke dissipated. Investigation of the condition found the motor starter for the Diesel Mixed Air Fan had failed. Prior to the start of the event, the HPCS system had been declared inoperable in accordance with plant Technical Specifications for planned maintenance.

The apparent cause of the motor starter failure was overheating of the contactor coil due to elevated system voltages. Corrective actions for this event include replacement of the contactor coil, increased frequency of preventative maintenance, and procedure revision. There were no other event-related equipment malfunctions.

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NARRATIVE

Plant Conditions
At the time of the event, the plant was operating in Mode 1 at 100% power. There were no structures, systems, or components that were inoperable at the start of the event that contributed to the event.

Event Description
On January 25, 2017 at 1836 PST, with the High Pressure Core Spray (HPCS) System [BG] declared inoperable in accordance with plant Technical Specifications for planned maintenance, the Diesel Mixed Air (DMA) Fan [FAN] was in normal operation with the fan running continuously, supplying air to the HPCS diesel room. The control room received a loss of power alarm for the fan's motor starter [MSTR], which was then followed by notification of smoke in the HPCS diesel room. The Operator at the motor control center [MCC] for the motor starter opened the disconnect and the smoke dissipated. Inspection of the motor starter found that the contactor coil [CNTR] had overheated.

This event is reportable as an event that could have prevented fulfillment of safety functions needed to mitigate the consequences of an accident per 10 CFR 50.73(a)(2)(v)(D), should the HPCS diesel have received a start signal during a design basis event. This condition was reported under 10 CFR 50.72(b)(3)(v)(D) via Event Notification #52510 for an event or condition that, at the time of discovery, could have prevented fulfillment of a safety function needed to mitigate the consequences of an accident; however, the Event Notification was retracted on February 3, 2017, based on the determination that the HPCS System had been declared inoperable prior to the smoke incident in accordance with plant Technical Specifications.

Immediate Corrective Actions
Operations personnel opened the disconnect for the affected motor starter, at which point the smoke dissipated.

Assessment of Safety Consequences
There were no actual safety consequences associated with this event, as there was no loss of safety function. There was no change in plant status or operating condition, and there was no risk to the public at any time due to this event. At the time of the event, work on the HPCS Service Water System was being performed in accordance with 10 CFR 50.65(a)(4) for managing risk. In emergency conditions, the HPCS System is used to provide core cooling and, during a loss of coolant accident, maintains reactor inventory. The DMA fan maintains suitable temperatures within the HPCS diesel generator room to ensure equipment operability temperature limits are not exceeded during an emergency. Had the HPCS System been in service and credited for accident mitigation, this event could have challenged the ability of the HPCS System to perform its safety function.

Cause of Event
The investigation into the apparent cause of the unexpected fan failure determined that the contactor coil for the fan's motor starter had overheated. The cause evaluation determined that the premature coil failure was likely caused by two contributing factors: increased coil operating currents and heating caused by elevated system voltages, and increased heating of the coil assembly caused by high magnet assembly temperatures due to loose shading coils.

LICENSEE EVENT REPORT (LER) CONTINUATION SHEET

(See NUREG-1022, R.3 for instruction and guidance for completing this form http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/)

1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER		
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NARRATIVE

Similar Events
A review of Corrective Action Program documents did not identify any previous starter coil failures for this particular manufacturer. Similar equipment from a different manufacturer recently experienced coil failures; Energy Northwest is working to identify the cause and develop a corrective action plan. Neither of the previous coil failures in the equipment has resulted in a loss of safety function or required an LER.

Further Corrective Actions
The contactor coil, magnet assembly, and upper contactor assembly were replaced. The starter was retested and returned to service. For critical starters from this manufacturer the frequency of the inspection maintenance will increase, and plant procedures will be revised to ensure adequate inspection of all magnet assemblies and coils.

Energy Industry Identification System (EIIS) Information codes from IEEE Standards 805-1984 and 803-1983 are represented in brackets as [XX] and [XXX] throughout the body of the narrative.